symbiOS: Minimal Sovereign Informational Kernel

Abstract

This paper presents symbiOS, a sovereign informational kernel designed for high coherence and security. It introduces the concepts of Mem-Nano-Bit (MNB) as a minimal unit of auditable semantic memory and COER (Cor-Nano-Bit) as a coherence engine governed by mathematical invariants (Ψ, Ω, CVaR, PoLE).

Core Architecture

The system is built on three sovereign layers:

1. **Scientific Layer**: Formal metrics and thermodynamic mirror concepts.
2. **Normative Layer**: Laws and invariants governing state transitions.
3. **Runtime Layer**: Executable environment for the kernel.

Security^2 (S^2)

Security^2 protects the security logic itself, metrics, and governance against internal corruption and semantic capture.

- **Metric Security**: Deterministic and auditable calculation of Ψ, Ω, CVaR.
- **Invariant Security**: Ω-GATE verification before any state write.
- **Thermodynamic Security**: Asymmetric cost for attackers vs. system adaptation.

Implementation

The kernel uses a 3-body persistent memory model. The state is represented geometrically as an organic cube where:

- **Purple/Violet**: High coherence (High Ψ).
- **Blue**: Normal operation.
- **Red**: Critical state.

Conclusion

symbiOS transforms coherence into infrastructure, providing a measurable, auditable, and publishable framework for autonomous organisms.

Institutional Closure (Ω-Closure)

The system transition from “architectural” to “institutional” is measured by the Closure Score ($F_{closure}$): $F_{closure} = (\text{genesis unique}) \wedge (\text{deterministic pipeline}) \wedge (\text{verifiable replay}) \wedge (\text{independent public evidence})$

Current state diagnosis: $F_{closure} \approx 0.6$. The system has high internal coherence but requires external verification.

Ω-Closure Validator

A formal module to evaluate the institutional state:

- **Genesis Integrity**: SHA-256 verification.
- **Merkle Root**: Causal consistency.
- **External Verification**: Audit from independent platforms (Hugging Face, Zenodo, Sepolia).

Conclusion (Updated)

symbiOS is now entering the “Institutional Phase”, moving beyond speculative design to certified, auditable existence.

Cognitive Memory Architecture (LTM/STM/Buffer)

The system transition from passive storage to an operational cognitive system:

- **LTM (Long-Term Memory)**: Persistent set of all MNBs $\{x \in M\}$.
- **STM (Short-Term Memory)**: Coherent subset $\{x \in LTM \mid \Psi(x) > \theta\}$.
- **Buffer**: Operational Top-K selection for immediate action.

Decision Scoring Function

To move beyond relevance to actual decision-making, the system implements: $ $Score(x) = \alpha\Psi(x) + \beta Recency(x) + \gamma Utility(x) Where$\gamma Utility(x)$ ensures the system optimizes for institutional goals, not just semantic similarity.

Genesis Release (v1.2)

This version marks the “Cognitive Phase”, establishing the foundation for a unique, irreversible, and verifiable Genesis event.